

Henrique Baptista

AI Engineer | Agents · LLM Evaluation · Voice AI

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PROFESSIONAL SUMMARY

AI Engineer who builds production agent, LLM, and Voice AI systems — and the evaluation layer that proves they work. Three years owning AI problems end to end across healthcare and SaaS: LLM-as-Judge evaluators in production, multi-agent systems with LangGraph, an MCP server, ASR benchmarking with domain metrics, and live products I design, ship, and operate myself.

PROFESSIONAL EXPERIENCE

ML / AI Engineer - VOA Health

Feb 2025 – Jul 2026

São Paulo, Brazil · HealthTech startup — AI for clinical documentation

The product generated clinical documents at scale, but quality was invisible — failures surfaced only when clinicians noticed. My role: make quality measurable and turn it into engineering decisions.

LLM Quality & Evaluation

- Shipped **LLM-as-Judge evaluators to production** (LangSmith online evaluators) scoring AI-generated clinical documents on faithfulness, correctness, and format adherence; iterated the judge pipeline from 8% uncertain verdicts to **0%** and raised correction-validator approval from 0% to **80%**
- Calibrated judges against human annotation with **Cohen's κ agreement gates** before rollout; replaced ad-hoc review with a QA gate combining deterministic detectors and a root-cause bug catalog

Voice AI & ASR

- Benchmarked **9 commercial ASR providers on 10k+ clinical audio samples** (WER/CER, latency, cost) and created a medication-error metric; built a taxonomy of **217 real medication transcription errors** that drove provider selection and normalization rules
- Cut medication transcription errors by **78%** and WER by **21%** with a domain prompt plus lexical-phonetic post-processor — statistically validated (Wilcoxon, $N=1,227$); tuned VAD for noisy consultation audio

Data Pipelines & Observability

- Built the fine-tuning data curation pipeline end to end: audio chunking, Hugging Face Hub datasets, Argilla annotation with **dual-annotator consensus and adjudication**, live progress dashboard, and a containerized 24/7 webhook service — zero manual steps from annotator agreement to training-ready dataset
- Instrumented the realtime transcription pipeline with **23 latency checkpoints** (mic to UI) exposed via Prometheus/Grafana; infrastructure analysis identified the serving endpoint with 133ms RTT and 5ms jitter

PROJECTS

WordinAI — solo-built, live at wordinai.com

LangGraph · deepagents · Next.js · Supabase

- SaaS turning long-form audio into structured documents via a **multi-agent LangGraph system** — context analyst, source verifier (every claim needs a reference), action strategist — over hybrid retrieval (vector + full-text) with untrusted-content isolation against prompt injection
- Operated end to end: multi-provider STT with circuit breakers (OpenAI, Groq, Stripe), in-browser Whisper WASM privacy mode, 70+ test files including agent contract evals, Sentry + PostHog observability

WebSkill — MCP server

Python · FastMCP · ChromaDB

- Model Context Protocol (MCP) server that installs whole websites as queryable skills for any LLM: crawl → chunk → embed → retrieve, with **content-hash diffing** so refreshes only re-embed changed pages; OpenAI or local embeddings; 56 tests

Education Risk Prediction — Datathon Passos Mágicos

XGBoost · FastAPI · Evidently AI

- Early-warning system for student risk at an NGO: XGBoost classifier (**92.59% accuracy, 0.97 ROC-AUC**), FastAPI service, decision dashboard, and drift monitoring, deployed on Hugging Face Spaces

Insurance Claim Computer Vision — Porto Seguro Challenge (Top-10 Finalist)

Detectron2 · Mask R-CNN · AWS

- Detectron2 segmentation pipeline with a COCO dataset built from scratch and transfer learning on AWS — cutting manual claim-review workload by ~80%

EDUCATION

Postgraduate Program, Machine Learning Engineering - FIAP

Completed Mar 2026

Bachelor, Systems Analysis & Development - FIAP

Completed Dec 2024

TECHNICAL SKILLS

- LLM & Agents:** LangGraph, LangChain, deepagents, OpenAI & Anthropic APIs, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), hybrid search, tool calling, structured output, prompt engineering & caching
- Evaluation:** LLM-as-Judge, LangSmith, evaluation datasets, WER/CER benchmarking, human-annotation consensus (Argilla), Cohen's κ calibration, failure analysis, production QA
- ML & Data:** PyTorch, Hugging Face (Transformers, Datasets, Hub), Whisper, fine-tuning data curation, XGBoost, computer vision (Detectron2, YOLO)
- Engineering:** Python, TypeScript, FastAPI, Next.js, PostgreSQL, Supabase, Docker, CI/CD, Git, AWS (S3, EC2), Prometheus/Grafana, drift monitoring (Evidently)
- Languages:** Portuguese (Native) · English (Fluent)